

Seiberg-Witten Theory and the Intersection Form

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Abstract

This is temporary abstract text. The purpose of this dummy document is to create enough structure to evaluate the table of contents, chapter ordering, section hierarchy, and later typography choices. The content is placeholder text and should be replaced by the final thesis abstract.

Contents

Abstract	1
1 Hypotheses	5
1.1 Research Hypotheses	5
1.2 Decision Support Assumptions	5
1.3 Risk and Uncertainty Assumptions	5
1.4 Point-in-Time Assumptions	6
2 Introduction	7
2.1 Motivation	7
2.2 Problem Formulation	7
2.3 Research Questions	7
2.4 Scope and Delimitations	8
2.5 Thesis Structure	8
3 Background	9
3.1 Sequential Decision-Making	9
3.2 Reinforcement Learning	9
3.3 Distributional Reinforcement Learning	9
3.4 Financial Decision Support Systems	10
3.5 Risk and Uncertainty	10
3.6 Point-in-Time Data	10
4 Related Work and State of the Art	11
4.1 Reinforcement Learning in Finance	11
4.2 FinRL and Financial Reinforcement Learning Frameworks	11
4.3 Distributional RL and Implicit Quantile Networks	12
4.4 Risk-Sensitive Reinforcement Learning	12
4.5 Explainable Decision Support	12
4.6 Auditability and Traceability	12
5 Methodology	14
5.1 Research Design	14
5.2 Data Pipeline	14
5.3 State Space	14
5.4 Action Space	15
5.5 Reward Design	15
5.6 Evaluation Design	15

6	Theory	16
6.1	Markov Decision Processes	16
6.2	Value Functions	16
6.3	Quantile Regression	16
6.4	Implicit Quantile Networks	17
6.5	Risk Measures	17
6.6	Decision Support Theory	17
7	Experiments	18
7.1	Experimental Setup	18
7.2	Baseline Models	18
7.3	Training Procedure	18
7.4	Backtesting Protocol	19
7.5	Metrics	19
7.6	Robustness Checks	19
8	Proof of Concept	20
8.1	System Overview	20
8.2	Application Architecture	20
8.3	Portfolio Builder	20
8.4	Strategy Configuration	21
8.5	Decision Engine	21
8.6	Audit Log	21
9	Results	22
9.1	Overview of Results	22
9.2	Decision Outputs	22
9.3	Risk Summaries	22
9.4	Backtest Results	23
9.5	Qualitative Evaluation	23
9.6	Limitations of Results	23
10	Conclusion	24
10.1	Summary of Findings	24
10.2	Answer to Research Questions	24
10.3	Contributions	24
10.4	Limitations	25
10.5	Final Remarks	25
11	Future Perspectives	26
11.1	Improved Reinforcement Learning Agents	26
11.2	Multi-Asset Portfolio Extensions	26
11.3	Improved Uncertainty Estimation	26
11.4	Deployment and Infrastructure	27
11.5	Human-in-the-Loop Evaluation	27
12	References	28
13	Bibliography	29

A	Additional Material	30
A.1	Experiment Configuration	30
A.2	Additional Tables	30

Chapter 1

Hypotheses

1.1 Research Hypotheses

The thesis begins from a set of working assumptions about reinforcement learning, investment decision support, and uncertainty communication.

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1.2 Decision Support Assumptions

A decision support system should assist the investor without replacing the investor as the final decision-maker.

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1.3 Risk and Uncertainty Assumptions

The quality of a recommendation depends not only on expected return but also on downside risk and uncertainty estimates.

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1.4 Point-in-Time Assumptions

The system should only use information available at the time of each decision.

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Chapter 2

Introduction

2.1 Motivation

Stock investors make sequential decisions under uncertainty. The motivation for this thesis is to explore how reinforcement learning can support such decisions in a transparent and risk-aware way.

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2.2 Problem Formulation

The central problem is how to design a stock investment decision support system that provides meaningful recommendations while avoiding look-ahead bias.

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2.3 Research Questions

The thesis investigates how reinforcement learning, distributional value estimation, and point-in-time data handling can be combined in a decision support system.

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2.4 Scope and Delimitations

The thesis focuses on a proof of concept rather than a production trading platform.

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2.5 Thesis Structure

The thesis is organized into background, related work, methodology, theory, experiments, proof of concept, results, conclusion, and future perspectives.

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Chapter 3

Background

3.1 Sequential Decision-Making

Sequential decision-making concerns situations where each decision changes the future state of the system.

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3.2 Reinforcement Learning

Reinforcement learning studies agents that interact with environments through states, actions, and rewards.

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3.3 Distributional Reinforcement Learning

Distributional reinforcement learning models the distribution of returns rather than only the expectation.

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3.4 Financial Decision Support Systems

A financial decision support system assists users by organizing information and presenting recommendations.

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3.5 Risk and Uncertainty

Risk and uncertainty are central to investment decision-making and motivate the use of quantiles and downside metrics.

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3.6 Point-in-Time Data

Point-in-time data handling ensures that the system does not use information from the future.

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Chapter 4

Related Work and State of the Art

4.1 Reinforcement Learning in Finance

Existing research applies reinforcement learning to portfolio optimization, trading, and market simulation.

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4.2 FinRL and Financial Reinforcement Learning Frameworks

FinRL is relevant because it provides financial reinforcement learning environments, workflows, and examples.

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4.3 Distributional RL and Implicit Quantile Networks

Implicit Quantile Networks estimate return distributions and are relevant for downside-aware recommendations.

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4.4 Risk-Sensitive Reinforcement Learning

Risk-sensitive reinforcement learning studies decision-making objectives that account for variance, tail risk, or other risk criteria.

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4.5 Explainable Decision Support

Explainability is important when recommendations are intended for human decision-makers.

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4.6 Auditability and Traceability

Auditability makes it possible to reconstruct decisions, inputs, and outputs after the fact.

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Chapter 5

Methodology

5.1 Research Design

The methodology combines system design, reinforcement learning experiments, and proof-of-concept implementation.

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5.2 Data Pipeline

The data pipeline ingests market data, stores processed data, and prepares point-in-time datasets.

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5.3 State Space

The state space represents market conditions, portfolio status, investor profile, and strategy settings.

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5.4 Action Space

The action space defines recommendations such as hold, buy, sell, reduce, re-balance, or switch strategy.

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5.5 Reward Design

The reward design balances return, transaction costs, risk exposure, and strategy alignment.

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5.6 Evaluation Design

Evaluation considers backtesting, baselines, risk metrics, and qualitative inspection of decision logs.

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Chapter 6

Theory

6.1 Markov Decision Processes

A Markov decision process formalizes sequential decisions through states, actions, transitions, and rewards.

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6.2 Value Functions

Value functions estimate the expected return obtained from a state or state-action pair.

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6.3 Quantile Regression

Quantile regression provides a way to estimate conditional quantiles of a distribution.

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6.4 Implicit Quantile Networks

Implicit Quantile Networks approximate quantile functions and can represent return distributions.

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6.5 Risk Measures

Risk measures such as Value-at-Risk and Conditional Value-at-Risk summarize downside exposure.

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6.6 Decision Support Theory

Decision support theory emphasizes the role of systems that assist rather than replace human judgment.

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Chapter 7

Experiments

7.1 Experimental Setup

The experimental setup defines datasets, tickers, time periods, baselines, and evaluation criteria.

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7.2 Baseline Models

Baseline models provide simple points of comparison such as buy-and-hold or naive portfolio allocation.

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7.3 Training Procedure

The training procedure describes how agents are trained, validated, and evaluated.

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7.4 Backtesting Protocol

The backtesting protocol defines how historical decisions are simulated without using future information.

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7.5 Metrics

Metrics include returns, drawdown, risk-adjusted performance, quantile summaries, and decision consistency.

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7.6 Robustness Checks

Robustness checks examine whether results change under alternative assumptions or parameter settings.

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Chapter 8

Proof of Concept

8.1 System Overview

The proof of concept demonstrates the decision support workflow from data ingestion to recommendation output.

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8.2 Application Architecture

The application architecture separates data storage, model logic, decision engine, and user interface.

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8.3 Portfolio Builder

The portfolio builder allows a user to define holdings, cash, tickers, and investment horizon.

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8.4 Strategy Configuration

Strategy configuration allows the user to select or define investment strategies and risk preferences.

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8.5 Decision Engine

The decision engine transforms model outputs into human-readable recommendations.

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8.6 Audit Log

The audit log records inputs, model outputs, recommendations, and user decisions.

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Chapter 9

Results

9.1 Overview of Results

The results summarize the behavior of the proof of concept and the experimental findings.

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9.2 Decision Outputs

Decision outputs include recommended actions, quantile estimates, downside summaries, and explanations.

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9.3 Risk Summaries

Risk summaries present downside risk and uncertainty estimates associated with recommendations.

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9.4 Backtest Results

Backtest results compare the decision support approach with simpler baselines.

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9.5 Qualitative Evaluation

Qualitative evaluation inspects whether recommendations are understandable and aligned with strategy settings.

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9.6 Limitations of Results

The results are interpreted in light of the proof-of-concept scope and data limitations.

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Chapter 10

Conclusion

10.1 Summary of Findings

This section summarizes the main findings of the thesis.

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10.2 Answer to Research Questions

This section answers the research questions based on the methodology and results.

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10.3 Contributions

This section summarizes the practical, methodological, and conceptual contributions.

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10.4 Limitations

This section describes limitations related to data, modelling, evaluation, and implementation.

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10.5 Final Remarks

This section closes the thesis with final reflections.

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Chapter 11

Future Perspectives

11.1 Improved Reinforcement Learning Agents

Future work may investigate stronger agents, richer environments, or longer training runs.

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11.2 Multi-Asset Portfolio Extensions

Future versions may support broader portfolio-level allocation and rebalancing.

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11.3 Improved Uncertainty Estimation

Future work may combine distributional RL with alternative uncertainty estimation methods.

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11.4 Deployment and Infrastructure

Future work may deploy the proof of concept as a containerized application.

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11.5 Human-in-the-Loop Evaluation

Future studies may evaluate the system with real users and decision-making scenarios.

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Chapter 12

References

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Chapter 13

Bibliography

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Appendix A

Additional Material

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A.1 Experiment Configuration

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A.2 Additional Tables

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